AS · Unit 1 · WPH11 Mechanics & Materials

Energy

Energy never appears or disappears; it only moves from one store to another. Work done measures how much energy a force transfers, power measures how quickly it happens, and efficiency tells you how much ended up where you wanted it. With these few ideas you can solve problems that would be painful with forces and SUVAT alone.

Work done

In physics, **work** has a precise meaning: a force does work when its point of application moves in the direction of the force. Holding a heavy bag still is tiring, but your force on the bag does no work on it, because the bag doesn't move. Lifting it does.

Work done by a force

$$W = Fs$$

W: work done / J F: force / N s: distance moved in the direction of the force / m

DEFINITION

Work done and the joule

Work done = force × distance moved in the direction of the force. **1 joule** is the work done when a force of 1 N moves its point of application 1 m in the direction of the force, so $1\ \text{J} = 1\ \text{N m}$.

The big idea that makes work useful: **work done = energy transferred**. When you push a trolley and do 50 J of work on it, 50 J of energy leaves your chemical store and goes into the trolley's kinetic store (and some into thermal energy, if there's friction).

When the force is at an angle

Pull a sledge with a rope at an angle and only part of your pull moves it forwards. The rest just lifts the front of the sledge a little. Resolve the force along and perpendicular to the motion:

Work done by a force at an angle θ to the motion

1. The force F acts at angle θ to the direction of motion. Resolve it into a component along the motion and one at right angles:

$$F_{\parallel} = F\cos\theta \qquad\qquad F_{\perp} = F\sin\theta$$

2. The perpendicular component does no work: the sledge doesn't move in that direction at all. Only the component along the motion does work over the distance s:

$$\boxed{W = Fs\cos\theta}$$

At $\theta = 0$ this is just Fs. At $\theta = 90^\circ$, $\cos\theta = 0$: a force at right angles to the motion does no work.

- **Zero work:** any force at right angles to the motion. The normal reaction on a car driving along a flat road, and the weight of a ball rolling across a level floor, do no work.
- **Negative work:** a force opposite to the motion ($\theta = 180^\circ$, $\cos\theta = -1$), like friction or drag. It takes energy away from the object, usually into thermal energy.

WORKED EXAMPLE

A child pulls a sledge 25 m across flat snow with a force of 60 N along a rope at 30° to the horizontal. How much work does the pull do on the sledge?

Show solution

$$W = Fs\cos\theta = 60 \times 25 \times \cos 30^\circ = 1300\ \text{J}\ (1.3\ \text{kJ})$$

If the sledge moves at a steady speed, its kinetic energy doesn't change, so all 1300 J is transferred to thermal energy by friction with the snow.

You do need the area idea for one case: the area under a force–extension graph is the elastic strain energy stored in a stretched material (see Stretching Materials).

Kinetic energy

A moving object has energy in its **kinetic store**. To find how much, ask how much work you would have to do to get it moving from rest.

Kinetic energy

$$E_k = \frac{1}{2}mv^2$$

E_k: kinetic energy / J m: mass / kg v: speed / m s^{-1}

Deriving $E_k = \frac{1}{2}mv^2$ from $W = Fs$

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 A constant resultant force F pushes a mass m, starting from rest, through a distance s. The work done on it is

$$W = Fs$$

2 Newton's second law gives $F = ma$, so

$$W = mas$$

3 The acceleration is constant, so use $v^2 = u^2 + 2as$ with $u = 0$:

$$v^2 = 2as \quad \Rightarrow \quad as = \frac{1}{2}v^2$$

4 Substitute for as. All the work done has gone into the kinetic store:

$$W = m\left(\frac{1}{2}v^2\right) \quad \Rightarrow \quad \boxed{E_k = \frac{1}{2}mv^2}$$

Why v^2? To reach a higher speed with the same force you must push for longer (time $\propto v$), and the object is also moving faster on average while you push (average speed $= \frac{1}{2}v$). The distance, and so the work, grows with both: $s = \frac{1}{2}v \times t \propto v^2$. That average speed is also where the ½ comes from.

The v^2 has real consequences. Double the speed and the kinetic energy goes up **four** times. To stop a car, the brakes must do work equal to its kinetic energy, $Fd = \frac{1}{2}mv^2$. With the same braking force, the braking distance is proportional to v^2: from 30 mph to 60 mph the braking distance doesn't double, it quadruples.

WORKED EXAMPLE

A car of mass 1200 kg travels at 20 m s⁻¹. The brakes can supply a constant braking force of 6000 N. (a) Calculate its kinetic energy. (b) Find its braking distance. (c) What would the braking distance be at 40 m s⁻¹?

Show solution

$$\text{(a)}\quad E_k = \tfrac{1}{2}(1200)(20)^2 = 2.4 \times 10^5 \text{ J}$$

$$\text{(b)}\quad Fd = E_k \quad \Rightarrow \quad d = \frac{2.4 \times 10^5}{6000} = 40 \text{ m}$$

(c) Double the speed gives four times the kinetic energy, so with the same force the distance is $4 \times 40 = 160$ m.

Gravitational potential energy

Lift something up and you do work against its weight. That energy isn't lost: it is stored in the **gravitational potential energy** store, ready to come back out as kinetic energy when the object falls.

Change in gravitational potential energy near the Earth's surface

$$\Delta E_{grav} = mg\Delta h$$

ΔE_{grav}: change in GPE / J m: mass / kg g = 9.81 N kg⁻¹ Δh: change in height / m

Deriving the change in GPE, ΔE = mgΔh, from W = Fs

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Lift a mass m at a steady speed, so its kinetic energy doesn't change. With no acceleration, your upward force exactly balances its weight:

$$F = mg$$

2. You move that force through a height Δh in its own direction, so the work done is

$$W = F\Delta h = mg\,\Delta h$$

3 All of that work is stored as gravitational potential energy:

$$\boxed{\Delta E_{grav} = mg\Delta h}$$

This assumes g is constant, which is true near the Earth's surface. Over large distances g falls off, and you need the field ideas in Unit 5.

- Only **changes** in GPE matter. You can choose any level as zero (the floor, the table top, the ground) as long as you stick with it.
- Δh is the **vertical** height change. Walk up a long gentle ramp or climb a short steep ladder to the same height and the GPE gain is identical: the ramp needs a smaller force over a longer distance.

WORKED EXAMPLE

A 65 kg climber climbs from a ledge 12 m up a cliff to the top, which is 30 m above the ground. How much gravitational potential energy does she gain?

Show solution

Only the change in height counts: $\Delta h = 30 - 12 = 18$ m.

$$\Delta E_{grav} = mg\Delta h = 65 \times 9.81 \times 18 = 1.15 \times 10^4 \text{ J}$$

Conservation of energy

DEFINITION

Principle of conservation of energy

Energy cannot be created or destroyed. It can only be transferred from one store to another, so the total energy of a closed system stays constant.

For mechanics problems this becomes a bookkeeping rule. As an object moves, energy shifts between its kinetic and gravitational stores, and any work done against friction or drag is transferred to thermal energy in the surroundings:

$$\text{loss in } E_{grav} = \text{gain in } E_k + \text{work done against resistive forces}$$

With no friction, all the GPE lost becomes KE: $mg\Delta h = \frac{1}{2}mv^2$, so $v = \sqrt{2g\Delta h}$. The mass cancels, which is why a heavy skater and a light skater released from the same height reach the bottom with the same speed. Notice that this method doesn't care about the **shape** of the track, only the change in height. That's what makes energy so powerful: you'd need calculus to do this with forces.

WORKED EXAMPLE

A roller-coaster car of mass 500 kg starts from rest at the top of a slope 32 m above the ground and descends to a point 12 m above the ground. (a) Ignoring friction, find its speed there. (b) Its actual speed there is 17.0 m s^{-1}. How much work was done against resistive forces? (c) The track between the two points is 120 m long. Find the average resistive force.

Show solution

(a) It falls $\Delta h = 20\ \text{m}$. All the GPE lost becomes KE:

$$\tfrac{1}{2}mv^2 = mg\Delta h \quad \Rightarrow \quad v = \sqrt{2 \times 9.81 \times 20} = 19.8\ \text{m s}^{-1}$$

(b) Now do the full energy bookkeeping:

$$\Delta E_{grav} = 500 \times 9.81 \times 20 = 98\,100\ \text{J}, \qquad E_k = \tfrac{1}{2} \times 500 \times 17.0^2 = 72\,250\ \text{J}$$

$$\text{work done against resistive forces} = 98\,100 - 72\,250 = 2.59 \times 10^4\ \text{J}$$

(c) Work done = force × distance, so

$$F = \frac{2.59 \times 10^4}{120} = 215\ \text{N}$$

Bouncing balls

A dropped ball turns GPE into KE on the way down. During the bounce it squashes, storing energy elastically, then springs back. But some energy is transferred to thermal energy in the ball and the floor, and to sound, so it rebounds with less KE and can't reach its starting height. The fraction of energy kept in each bounce is the same as the ratio of rebound height to drop height.

WORKED EXAMPLE

A tennis ball of mass 0.058 kg is dropped from 2.0 m and rebounds to 1.3 m. (a) Find its speed just before it hits the ground. (b) How much energy is transferred to other stores in the bounce? (c) What percentage of its energy does it keep?

Show solution

$$\text{(a)}\quad v = \sqrt{2g\Delta h} = \sqrt{2 \times 9.81 \times 2.0} = 6.3 \text{ m s}^{-1}$$

$$\text{(b)}\quad mg(2.0 - 1.3) = 0.058 \times 9.81 \times 0.70 = 0.40 \text{ J}$$

(c) Its GPE at the top of each bounce is proportional to the height, so it keeps $1.3/2.0 = 0.65$, or 65%, of its energy.

COMMON MISTAKE

Energy is never "used up" or "lost" in the sense of disappearing. When a question says "energy lost", it means energy transferred to stores you can't easily use, usually thermal energy of the surroundings. Your final answer should still balance: energy in = energy out.

Power

Two cranes can lift the same load to the same height, doing the same work. The more powerful one does it faster. **Power** is the rate of transferring energy, or the rate of doing work.

Power from energy transferred

$$P = \frac{E}{t}$$

P: power / W E: energy transferred / J
t: time / s

Power from work done

$$P = \frac{W}{t}$$

W: work done / J t: time / s

DEFINITION

Power and the watt

Power is the rate of energy transfer (or the rate of doing work). **1 watt** is a rate of energy transfer of 1 joule per second: $1 \text{ W} = 1 \text{ J s}^{-1}$.

COMMON MISTAKE

Don't confuse the symbols: W for work done (in J) and W for the watt (the unit of power). In $P = W/t$, the W is work. And watch for kW: 1 kW = 1000 W.

WORKED EXAMPLE

A crane lifts a load of mass 800 kg through a height of 15 m in 20 s at a steady speed. (a) Calculate the useful power output. (b) The motor's input power is 9.0 kW. Calculate its efficiency.

Show solution

(a) At steady speed, the useful work done is the gain in GPE:

$$P = \frac{mg\Delta h}{t} = \frac{800 \times 9.81 \times 15}{20} = 5890 \text{ W } (5.9 \text{ kW})$$

$$\text{(b)} \quad \text{efficiency} = \frac{\text{useful power output}}{\text{total power input}} = \frac{5890}{9000} = 0.65 = 65\%$$

Efficiency

No machine transfers all its input energy to where you want it. A motor heats up, its bearings rub, it hums. **Efficiency** measures the fraction of the input that is useful.

Efficiency (energy)

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$$

Efficiency (power)

$$\text{efficiency} = \frac{\text{useful power output}}{\text{total power input}}$$

The two forms are the same thing: divide the top and bottom of the energy form by the same time t and you get the power form. Efficiency has no units. Give it as a decimal (0.65) or a percentage (65%), and it can never be more than 1, because that would mean creating energy. A **Sankey diagram** shows the same information as a picture: the width of each arrow is proportional to the energy or power it carries.

EXAM TIP

In efficiency questions, first decide what is **useful**. For a motor lifting a load it is the GPE gained, $mg\Delta h$; for a kettle it is the thermal energy of the water; for a light bulb it is the light. Then make sure the top and bottom of the fraction are both energies or both powers, never one of each.

Quick checks

QUICK CHECK

A waiter carries a tray across a room at a constant velocity, holding it level. How much work does the upward force of his hand do on the tray?

- It equals the tray's weight × the distance walked.
- Zero
- It equals the tray's kinetic energy.
- It depends on how fast he walks.

The force is vertical and the displacement is horizontal, so $\theta = 90^\circ$ and $W = Fs\cos 90^\circ = 0$. The tray's energy doesn't change: it stays at the same height and the same speed.

QUICK CHECK

A car's speed increases from 15 m s^{-1} to 30 m s^{-1}. With the same braking force, its braking distance becomes:

- the same
- twice as long
- four times as long
- eight times as long

Braking distance $d = \frac{1}{2}mv^2/F$. With m and F unchanged, $d \propto v^2$, so doubling v multiplies d by 4.

QUICK CHECK

A 1 kg ball and a 5 kg ball are released from rest at the same height on two frictionless slopes of different steepness. At the bottom, which is true?

- The 5 kg ball is faster.
- The ball on the steeper slope is faster.
- **They have the same speed.**
- The 1 kg ball is faster.

$mg\Delta h = \frac{1}{2}mv^2$ gives $v = \sqrt{2g\Delta h}$: the mass cancels and only the height change matters, not the shape of the slope. (The 5 kg ball does have five times the kinetic energy.)

QUICK CHECK

Which of these is equivalent to the watt in SI base units?

- kg m s^{-2}
- kg m^2 s^{-2}
- **kg m^2 s^{-3}**
- J s

1 W = 1 J s^{-1}, and 1 J = 1 N m = 1 kg m^2 s^{-2}. Dividing by seconds gives kg m^2 s^{-3}. The first option is the newton and the second is the joule.

QUICK CHECK

An electric motor with an input power of 40 W lifts a 2.0 kg mass through 3.0 m in 5.0 s. What is its efficiency?

- 15%
- 29%
- 68%
- 147%

Useful energy = $mg\Delta h = 2.0 \times 9.81 \times 3.0 = 58.9$ J. Energy input = $Pt = 40 \times 5.0 = 200$ J. Efficiency = 58.9 / 200 = 0.29. Comparing 58.9 J with 40 W gives the impossible 147%: never mix energy and power.

QUICK CHECK

A skater on a ramp with friction rises to a lower height on each swing. Where has the "missing" energy gone?

- It has been destroyed by friction.
- Mostly to thermal energy of the wheels, the ramp and the air.
- It has been converted into extra gravitational potential energy.
- It has been stored as kinetic energy in the ramp.

Work done against friction and drag transfers energy to thermal stores (and a little to sound). The total is conserved; it is just in a less useful form.